

ПОЛНАЯ ИСПРАВЛЕННАЯ КОНФИГУРАЦИЯ ПО МЕТОДИЧКЕ СИСА 2026

1. ISP

```
hostnamectl set-hostname isp
exec bash

# WAN (магистраль)
mcedit /etc/net/ifaces/ens3/options
TYPE=eth
BOOTPROTO=dhcp

# HQ
mkdir -p /etc/net/ifaces/ens4
echo "172.16.1.1/28" > /etc/net/ifaces/ens4/ipv4address
mcedit /etc/net/ifaces/ens4/options
TYPE=eth
BOOTPROTO=static

# BR
mkdir -p /etc/net/ifaces/ens5
echo "172.16.2.1/28" > /etc/net/ifaces/ens5/ipv4address
mcedit /etc/net/ifaces/ens5/options
TYPE=eth
BOOTPROTO=static

# IP Forward
mcedit /etc/net/sysctl.conf
net.ipv4.ip_forward = 1

systemctl restart network

apt-get update && apt-get install iptables -y

iptables -t nat -A POSTROUTING -s 172.16.1.0/28 -o ens3 -j MASQUERADE
iptables -t nat -A POSTROUTING -s 172.16.2.0/28 -o ens3 -j MASQUERADE

iptables-save > /etc/sysconfig/iptables
systemctl enable --now iptables

timedatectl set-timezone Europe/Moscow
```

2. HQ-RTR

```
enable
configure terminal

hostname HQ-RTR
ip domain-name au-team.irpo

username net_admin
password P@ssw0rd
role admin

interface isp
ip address 172.16.1.2/28
ip nat outside
exit

interface vl100
ip address 10.10.100.1/27
ip nat inside
exit

interface vl200
ip address 10.10.200.1/27
ip nat inside
exit

interface vl999
ip address 10.10.30.1/29
ip nat inside
exit

port gel
service-instance gel/vl100
encapsulation dot1q 100 exact
rewrite pop 1
connect ip interface vl100
exit
service-instance gel/vl200
encapsulation dot1q 200 exact
rewrite pop 1
connect ip interface vl200
exit
service-instance gel/vl999
encapsulation dot1q 999 exact
rewrite pop 1
connect ip interface vl999
exit

interface tunnel.0
```

```
description GRE-to-BR-RTR ip address
10.10.10.1/30 ip tunnel 172.16.1.2
172.16.2.2 mode gre ip ospf
authentication message-digest ip ospf
message-digest-key 1 md5 P@ssw0rd exit
```

```
router ospf 1
ospf router-id 10.10.10.1
passive-interface default no
passive-interface tunnel.0
network 10.10.100.0/27 area
0 network 10.10.200.0/27
area 0 network 10.10.30.0/29
area 0 network 10.10.10.0/30
area 0 exit
```

```
ip nat pool HQ 10.10.30.0-
10.10.200.255
```

```
ip nat source dynamic inside-
to-outside pool HQ overload
interface ISP
```

```
no dhcp-server 1
ip pool VLAN200 10.10.200.2-10.10.200.30
```

```
dhcp-server 1
pool VLAN200 1
mask 255.255.255.224
gateway 10.10.200.1 dns
10.10.100.2 domain-name
au-team.irpo
```

```
interface vl200
dhcp-server 1
exit
```

```
ntp timezone utc+3
write memory
```

3. HQ-SRV (DNS/BIND9)

```
hostnamectl set-hostname hq-srv.au-team.irpo

mkdir -p /etc/net/ifaces/ens3
echo "10.10.100.2/27" > /etc/net/ifaces/ens3/ipv4address
echo "default via 10.10.100.1" > /etc/net/ifaces/ens3/ipv4route

mcedit /etc/net/ifaces/ens3/options
TYPE=eth
BOOTPROTO=static

systemctl restart network

useradd sshuser -u 2026 echo
'sshuser:P@ssw0rd' | chpasswd
gpasswd -a sshuser wheel

mcedit /etc/sudoers %wheel
ALL=(ALL:ALL) NOPASSWD: ALL

mcedit /etc/openssh/sshd_config
Port 2026
MaxAuthTries 2
Banner /etc/openssh/banner
AllowUsers sshuser

echo "Authorized access only" > /etc/openssh/banner

systemctl restart sshd apt-get update && apt-get

install bind bind-utils -y

mcedit /var/lib/bind/etc/options.conf
listen-on port 53 { 10.10.100.2; any; };
forwarders { 77.88.8.8; }; allow-query {
any; };

mcedit /var/lib/bind/etc/rfc1912.conf
zone "au-team.irpo" { type master;
file "au-team.irpo";
};
zone "100.10.10.in-addr.arpa" { type
master;
file "10.10.100.in-addr.arpa";
};
zone "200.10.10.in-addr.arpa" { type
master;
file "10.10.200.in-addr.arpa";
};

cd /var/lib/bind/etc/zone cp
empty au-team.irpo cp empty
10.10.100.in-addr.arpa cp empty
10.10.200.in-addr.arpa
```

```

# ПРЯМАЯ ЗОНА
mcedit au-team.irpo
$TTL 1D
@ IN SOA hq-srv.au-team.irpo. root.au-team.irpo. (
2026051001 12H 1H 1W 1H )
@          IN NS          hq-srv.au-team.irpo.
hq-rtr IN A 10.10.100.1 br-rtr
IN A 10.20.20.1 hq-srv IN A
      10.10.100.2 hq-cli IN A
      10.10.200.2

br-srv IN A 10.20.20.2 docker IN
A      172.16.1.1 web          IN A
      172.16.2.1

# ОБРАТНАЯ VLAN100 mcedit
10.10.100.in-addr.arpa 1 IN
PTR hq-rtr.au-team.irpo.
2 IN PTR hq-srv.au-team.irpo.

# ОБРАТНАЯ VLAN200 mcedit
10.10.200.in-addr.arpa 1 IN
PTR hq-rtr.au-team.irpo. 2 IN
PTR hq-cli.au-team.irpo.

rndc-confgen > /etc/bind/rndc.key sed -i
'6,$d' /etc/bind/rndc.key chown -R
root:named /var/lib/bind/etc/zone/*

named-checkconf named-
checkconf -z systemctl
enable --now bind

echo "search au-team.irpo" > /etc/resolv.conf
echo "nameserver 10.10.100.2" >> /etc/resolv.conf

timedatectl set-timezone Europe/Moscow

```

4. HQ-CLI

```
hostnamectl set-hostname hq-cli.au-team.irpo

mkdir -p /etc/net/ifaces/ens3

mcedit /etc/net/ifaces/ens3/options
TYPE=eth
BOOTPROTO=dhcp

echo "search au-team.irpo" > /etc/resolv.conf
echo "nameserver 10.10.100.2" >> /etc/resolv.conf

systemctl restart network
timedatectl set-timezone Europe/Moscow
```

5. BR-RTR

```
enable
configure terminal

hostname BR-RTR
ip domain-name au-team.irpo

username net_admin
password P@ssw0rd
role admin

interface isp
ip address 172.16.2.2/28
ip nat outside
exit

interface br
ip address 10.20.20.1/28
ip nat inside
exit

port tel
service-instance tel/br
encapsulation untagged
connect ip interface br
exit

interface tunnel.0
description GRE-to-HQ-RTR
ip address 10.10.10.2/30
ip tunnel 172.16.2.2 172.16.1.2 mode gre
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 P@ssw0rd
exit

router ospf 1
ospf router-id 10.10.10.2
passive-interface default
no passive-interface tunnel.0
network 10.20.20.0/28 area 0
network 10.10.10.0/30 area 0
exit

ip nat source dynamic inside-to-outside interface isp overload

ip route 0.0.0.0/0 172.16.2.1

ntp timezone utc+3
write memory
```

```
ip nat pool BR-POOL 172.16.2.2 172.16.2.2ssss
```

```
ip nat source dynamic inside-to-outside pool BR-POOL overload interface isp
```

6. BR-SRV

```
hostnamectl set-hostname br-srv.au-team.irpo

mkdir -p /etc/net/ifaces/ens3
echo "10.20.20.2/28" > /etc/net/ifaces/ens3/ipv4address
echo "default via 10.20.20.1" > /etc/net/ifaces/ens3/ipv4route

mcedit /etc/net/ifaces/ens3/options
TYPE=eth
BOOTPROTO=static

useradd sshuser -u 2026
echo 'sshuser:P@ssw0rd' | chpasswd
gpasswd -a sshuser wheel

echo "search au-team.irpo" > /etc/resolv.conf
echo "nameserver 10.10.100.2" >> /etc/resolv.conf

systemctl restart network
timedatectl set-timezone Europe/Moscow
```